Gender in Pool … The "Battle of the Trans Sexes"

Dr. Dave Alciatore, PhD

Harriet Haynes, a professional English pool player has had an impressive tournament record over the years (see the sidebar). However, many of her competitors have not been happy with her success. For example, Lynne Pinches (in the English Pool Association's 2023 Champion of Champions) and Kim O'Brien (in the 2024 European Pool Championship) made it to the tournament Finals with Harriet and forfeited their matches just before the opening lag shot. Why would top women pros do such a thing, refusing the chance to win a prestigious tournament title? Well, Harriet is a transgender woman, and her opponents claim she has an unfair advantage against biological females. A transgender woman is a person born male who identifies as a female and often undergoes gender reassignment hormone treatments and elective surgeries to become a woman. Harriet's opponents forfeited their matches in protest to convince tournament organizations to change their transgender inclusion policies.

Harriet Haynes
English Pool Association
Women's Championships

- World Ladies Team Champions, 2024, 2022
- World Ladies Masters Champion, 2024, 2023
- European Ladies Champion, 2024
- World Scotch Doubles Champion, 2023
- European Champion, 2023
- European Team Champions, 2023, 2022, 2020
- National Ladies Singles, Champion 2022
- World Champion, 2022 (IPA)
- National Champion of Champions, 2021, 2018

Harriet Haynes, English Pool professional

In response to the controversy, the English Blackball Pool Federation (EBPF) decided to institute a transgender exclusion policy in female-only events. Here is a quote from their public statement: "*… only biological [females] would be eligible to compete in the Female Tour. We made this decision in the interests of fairness, because we believe that people who have gone through male puberty have a competitive advantage over biological [females].*" Under this new policy, Harriet will no longer be eligible to compete in Female Tour events. Instead, she will be required to enter Open events (with the men), as she did before she underwent transgender hormone therapy and surgeries to become a

woman. Harriet does not think it is fair to be excluded from female events since she considers herself a female and is legally recognized as such. She feels her exclusion is discrimination. She does not want to be forced to compete in Open divisions since she is legally female. In fact, she feels so strongly about this, she filed a lawsuit against the EBPF. In online video *cdn.jwplayer.com/previews/wfp9y1Ka*, Harriet explains the story in her own words and talks about the purpose of the lawsuit. What do you think? Should transgender women be allowed to compete in female cue sport events?

Harriet Haynes awarded championship trophy after Lynne Pinches forfeited

One thing that makes this legal case interesting is that transgender athletes have been in the news a lot lately with high profile stories. The most recent example involves Harriet making it into an Ultimate Pool Women's Pro Series Final against another Trans-Gender athlete (Lucy Smith). Harriet won the Final on April 6, 2025, the day before the EBPF discrimination trial began! Another is from NCAA women's volleyball, where some teams recently refused to play against the San Jose State team because it includes a transgender athlete. The NCAA does not currently exclude transgender athletes from women's events and this policy recently survived a court challenge, so the transgender volleyball player is officially allowed to continue to play, despite opponent team protests. Although, President Donald Trump recently signed an executive order (February 2025) aiming to ban transgender athletes from competing on girls' and women's sports teams by denying federal funds to any schools that allow it. The ramifications of this will probably take years to settle in courts or in Congress.

Another thing that makes Harriet's case interesting is that it is not obvious to most people that males should have an advantage over females in pool, except maybe with a power shot like the break. I personally have a strong interest in this potentially landmark legal case, not only because it is controversial and could set a precedent throughout all cue sports, but because I was asked to serve as an expert witness in support of the EBPF. My role, along with a biology expert, was to help convince a judge that being born male and having gone through puberty gives a transgender woman athlete an unfair advantage in competing against biological females in pool.

Billy Jean King and Bobby Riggs in a press conference before their "Battle of the Sexes" match

When I first heard about the case, two things came to mind. The first was the 1973 "Battle of the Sexes" tennis match between Billy Jean King and Bobby Riggs (see the photo above). The event was a media sensation and fun to watch, but all it proved was that one of the greatest professional female tennis players of all time could defeat a mediocre male pro of the time. The second thing that came to mind was the series of Bud Light "Ladies Night" TV commercials in the mid-1990s. The photo below is from one of the ads featuring a Ladies Night pool tournament where some men who really wanted to win some free Bud Light dressed up for the part. The photo shows them dramatically protesting after somebody harmlessly referred to them as "guys" ("Who you callin' guys?!"). A message of the ads, other than humor, was: *A man in lipstick and a dress still plays like a man.* This is not an appropriate analogy for the current transgender eligibility situation, but the message that males should not be allowed to participate in female-only pool events is certainly valid.

Bud Light "Ladies Night" TV commercial

Transgender Policies

Since the lawsuit was the result of a new transgender policy of the EBPF, I researched other current policies in the pool world. I could not find anything online for the World Pool Association (WPA), the international governing body for pocket billiards, but Billiards Digest reached out to the WPA and received a copy of their current draft policy (see below). I also researched the popular pool league systems. The APA currently has an official written policy indicating that the gender listed on any official government photo ID is used to determine eligibility (see the sidebar). I could not find any policies online for other league/tournament systems (CSI, BCAPL, VNEA, etc.). I would suspect that informally, if an issue arose at a tournament, the organizers would probably follow an APA-like policy. Therefore, if a transgender woman has their gender changed on their driver's license from male to female, which is not very difficult to do in many US States, she would be allowed to enter these league-system female tournaments.

American Poolplayers Association (APA)
Gender Eligibility Policy

Participation in some divisions, and events, is limited to members of certain genders. If the eligibility of a participant in one of these divisions, or events, is questioned, the gender listed on the State/Province issued picture ID, Military ID or passport will be used to determine eligibility.

World Pool Association (WPA)
Current Draft of the Eligibility Policy for Transgender Athletes

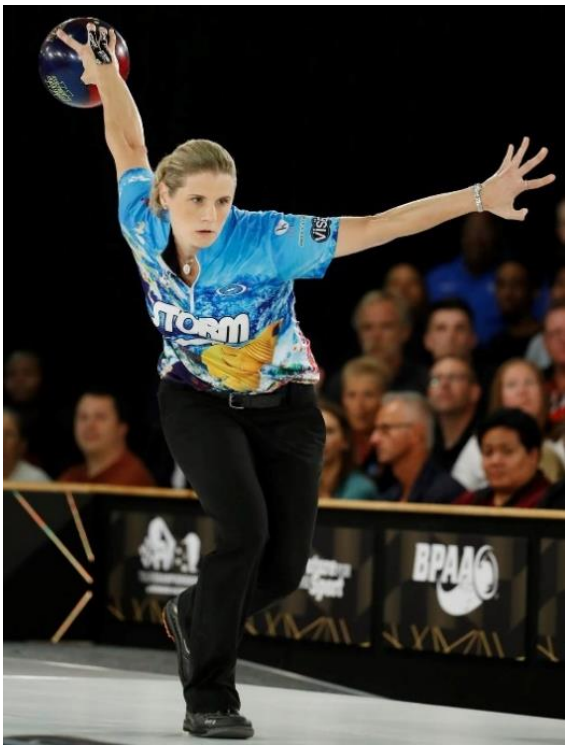
Transgender Female Athletes (Male-at-Birth to Female)

- Must submit a written declaration of female gender identity.
- Must continuously demonstrate serum testosterone levels below 10 nmol/L for at least 12 months before competition. Hyperandrogenism policies address cases where female athletes naturally have higher testosterone levels. Hormone testing is conducted when questions arise about an athlete's sex, not as universal screening.
- Additional hormonal parameters, such as estradiol and progesterone, may be evaluated to ensure compliance.
- Must not have competed in the male category internationally in the last four years.
- Ongoing compliance with these conditions is required.

General Provisions for All Transgender Athletes

- Specific requirements for transgender athletes, including surgical anatomical changes, legal recognition of their assigned sex, and a minimum period of hormonal therapy.
- Athletes switching categories may only do so after a waiting period of four years and compliance with the eligibility criteria for the new category.
- All participants must meet WPA anti-doping standards and competition rules.

I also researched the policy in bowling, thinking the issues might be similar. Although, compared to pool, strength seems to be more important in bowling since fast ball speed with lots of ball spin can provide an advantage. However, there have been some noteworthy tournament successes from female bowlers in events with males. Woman pro Kelly Kulick won the 2010 Professional Bowlers Association (PBA) Tournament of Champions against all the top men bowlers of the time. She was the first and currently only woman to ever win a PBA tour title. What made the victory even more special was that the tournament was a Major. Also, Gianna Brandolino, a 17-year-old junior woman bowler, won the 2024 PBA LBC National Championship Clash, where she outlasted men and women amateur, professional, and senior championship finalists in a roll-off format, winning the championship trophy and $60,000! Not bad for a petite teenage girl! A woman pool player has never won or even gone deep in a major open professional tournament.



Professional bowler Kelly Kulick and junior bowler Gianna Brandolino

I found a very thorough policy online from the United States Bowling Congress (USBC) concerning transgender women participating in elite events (see below). Measured testosterone levels must be below certain thresholds, there must be a statement from a medical doctor that birth gender does not offer a competitive advantage, and USBC staff must evaluate and determine that the biomechanics of the person's shot delivery (ball rev rate and speed for a given ball weight) are in typical female ranges. The second bullet near the end of the quoted policy is particularly interesting. It basically implies that if a transgender female bowler throws the ball like a male, she will not be allowed to compete against the females. The USBC, the WPA, and all governing bodies in sports will probably need to continue to modify and improve these policies over time as laws, legal precedents, presidential executive orders, and social customs change over the years.

United States Bowling Congress
Gender Classification Policy Excerpt

Elite female competition conditions:

1. The athlete has declared that her gender identity is female. The declaration cannot be changed, for sporting purposes, for a minimum of four years.
2. The athlete must demonstrate that her total testosterone level in serum has been below 10 nmol/L for at least 12 months prior to her first competition.
3. The athlete's total testosterone level in serum must remain below 10 nmol/L throughout the period of desired eligibility to compete in the female category.
4. Compliance with these conditions may be monitored by testing. In the event of non-compliance, the athlete's eligibility for female competition will be suspended for 12 months.
5. Athlete must submit declaration from a medical doctor stating to a reasonable degree of medical certainty that her birth gender does not give her a competitive advantage.
6. Athlete must complete an evaluation session at a training facility to capture video of athlete's delivery and ball tracking data. Video and data must be sent to USBC for review.

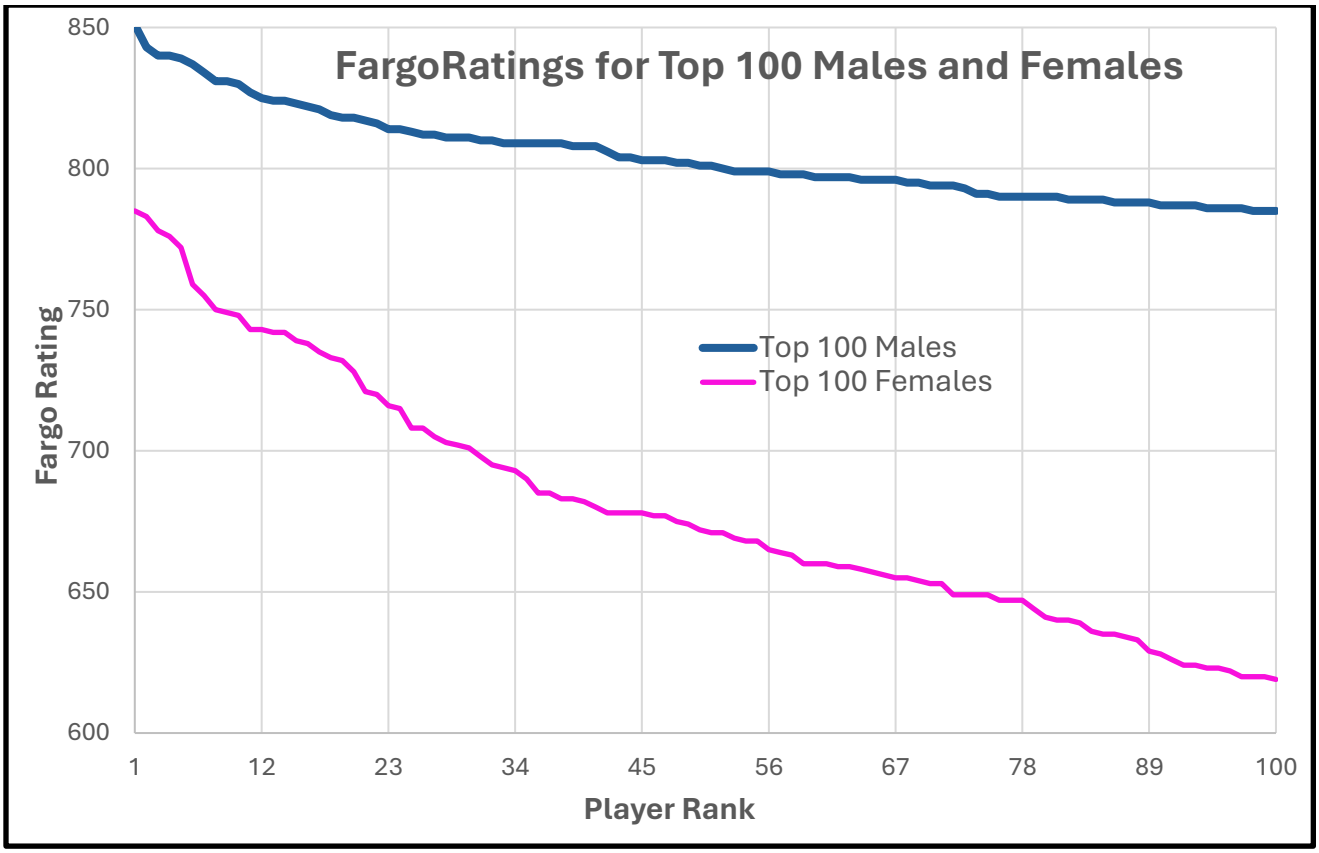
USBC Rules staff will evaluate the submission and make an administrative determination as to whether the application is approved or if the athlete's birth gender gives her a competitive advantage and the application is denied. Rules staff may request independent opinions from medical doctors, USBC trained coaches or other experts in making a determination. A determination will be based upon the following criteria:

- Evaluation of whether the biomechanics of the applicant's shot delivery give a competitive advantage due to birth gender.
- Evaluation of whether the applicant's ball tracking data falls within a reasonable range of elite female competition.
    - For example, if the applicant's rev rate and ball speed are more comparable to elite male athletes than female athletes, the application may be denied.
- Opinions of medical doctors.

Are Men Better at Pool Than Women?

Pool has been a male dominated sport over its entire history, and it remains that way now. Currently, many more males than females (about a 6 to 1 ratio) actively play the sport. As a result, there are many more top male players than female. Below are the FargoRatings (from *www.fargorate.com*) of the top 100 males and top 100 females in the world (as of September 2024). Notable features of the data include:

- There are no females in the top 100 of world ratings.
- The lower-ranked females in the top 100 (in the 620 range) are rated well below the lower-ranked males in the top 100 (in the 780 range). The difference between 780 and 620 (140) represents a very large difference in playing ability. The FargoRate system uses a logarithmic scale, so a gap of 100 points (at any level) predicts a 2-to-1 game-win ratio. Therefore, the 140-point rating difference means the 100th best elite male in the world is more than twice as good as the 100th best female!

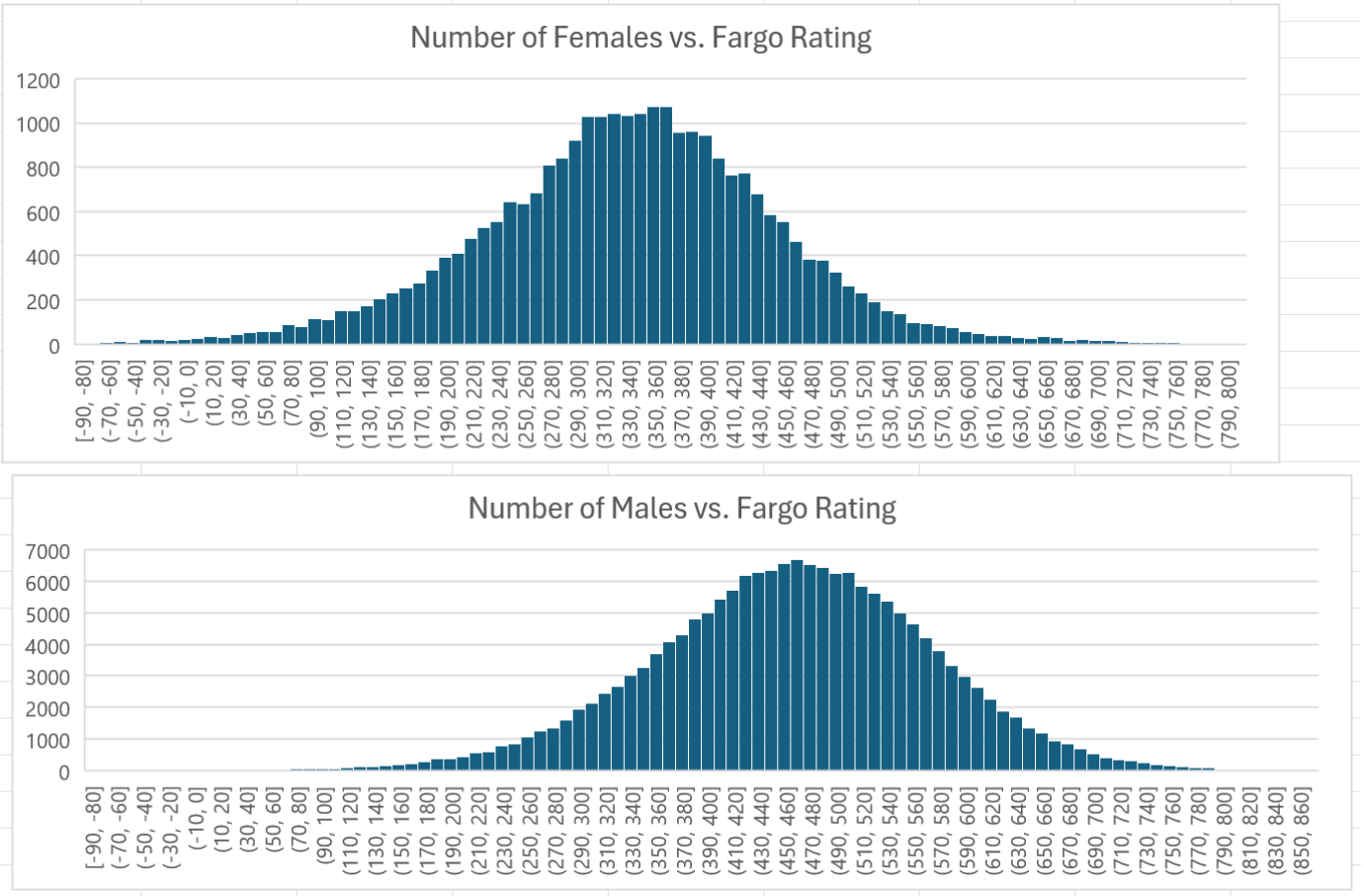


Top 100 Males and Females in FargoRate

Recently, Mike Page (co-founder of FargoRate) graciously agreed to send me an anonymous list of more than 200,000 FargoRatings (with genders) worldwide, including only established players with 50 or more games in the system as of December 2024. Thanks, Mike! This allowed me to do some analysis of gender distributions, as shown in the plots below. The first plot is called a histogram or bar plot. The horizontal axis shows ranges of FargoRatings, and the vertical axis shows the number of players in each FargoRate range. Per the plots of both females and males in the system, the current amateur male population clearly plays at a higher level than the current amateur female population. The average rating of the males is 460 compared to 337 for the females, with a difference of 123. So, the average male in the system is more than twice as good as the average female!

FargoRate includes worldwide game-result data from many pro pool tournaments, but the bulk of the data is from amateur pool league systems, which include amateur male and female players of all ability levels. There are more males in the system simply because many more males than females play pool, but there are currently still over 27,000

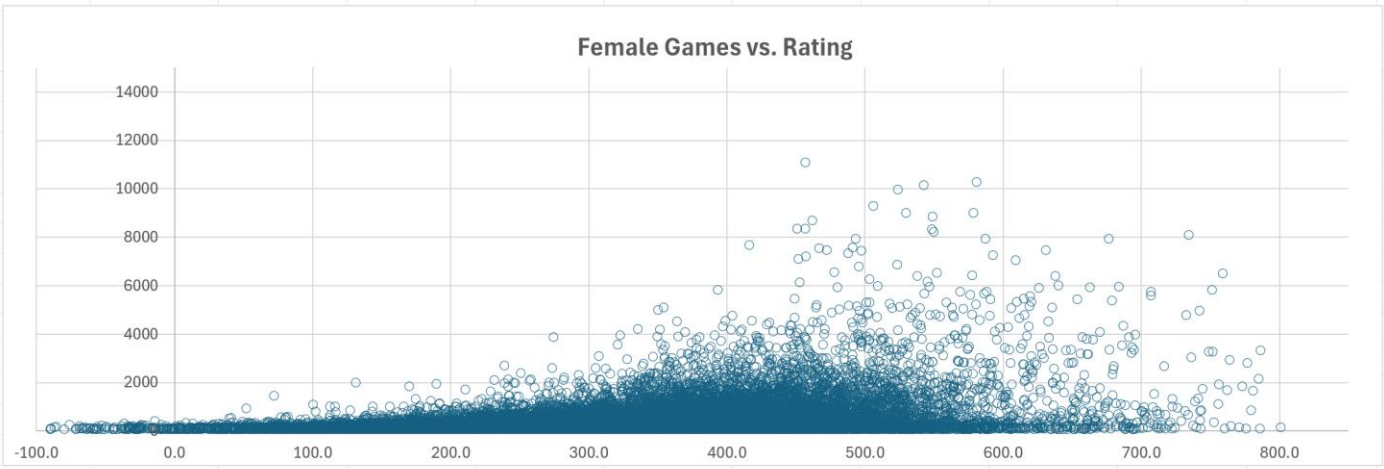
females in FargoRate. The rankings and rating distributions do not necessarily prove that females cannot play as well as males, but the data do clearly show that **females currently do not play at nearly the same level as males**. If there were many more females playing pool, there would likely be more females at the elite level. But what remains to be seen or known is: Would the average female be just as good as the average male if there were equal numbers of males and females playing? Based on the analysis and discussion in the remainder of this article, the answer is probably no, but we cannot know for sure without many more females participating in the sport.

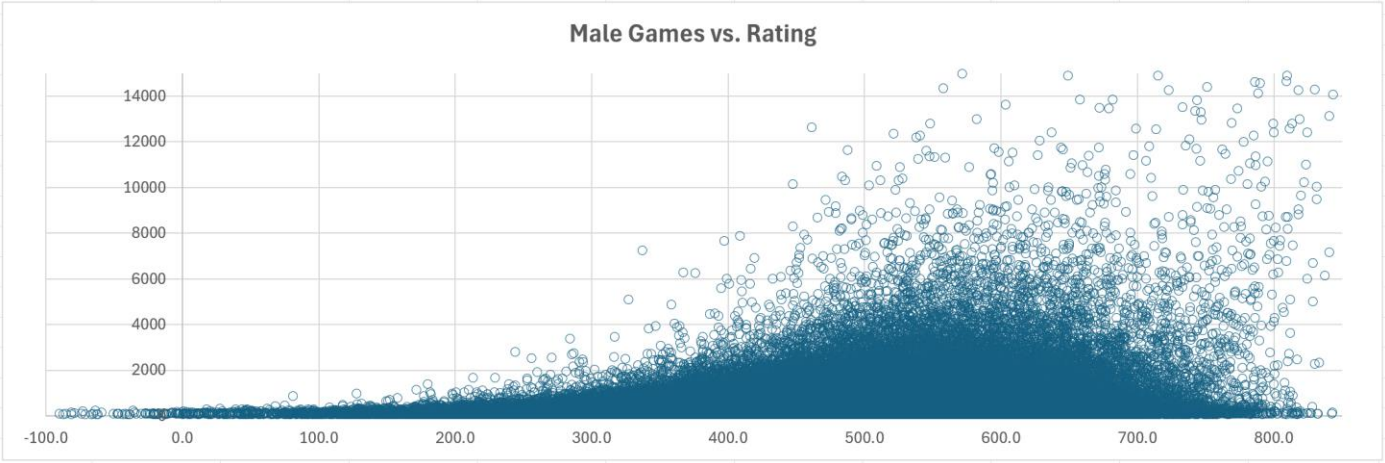


FargoRate Gender Distributions

The scatter plots below show a small blue circle for each person in the FargoRate system. The dark blue sections of the plots are due to many of the blue circles overlapping. Again, the horizontal axis is the FargoRating, but the vertical axis now shows the number of games each person has in the system. There is a very wide spread in the number of the games, but the median (middle) values for males (199) and females (180) are similar. The horizontal shift in the distributions relative to each other is expected due to the FargoRatings being higher for males than females.

In general, many males, especially at higher FargoRatings, have a larger number of games in the system than most females, but this is simply because they are playing in more leagues or tournaments. Better players generally play competitively more often because they have a better chance to do well in competition, and they also like to challenge themselves against other competitive players. There are also fewer female games, even at the top level, partly because there are many more Open (Men's) tournaments than Women's tournaments. Also, even when females enter Open tournaments, they do not go as deep or play as many matches since they are not as competitive as the males.





Number of Games a Different FargoRatings

It is also interesting to look at players that fall within different ranges of game numbers, even though the number of games in the system does not necessarily correlate with ability level. One interesting category of players is those with only 200-500 games in the system. We could generally classify them as “recreational;” although, there are also some elite players in this range. The average female rating in the 200-500 game range is 342, which is much lower (by 121) than the average male rating of 463. So apparently, “recreational” males also play much better than “recreational” females. Another interesting category is “experienced” players with more than 3000 games in the system. The average female rating in this category is 512, which is still lower (by 76) than the average male rating of 588. Again, “experienced” males also play much better than “experienced” females; but not by as wide a margin as “recreational” players.

Caution should be exercised when making conclusions based on the numbers of games in the FargoRate system. It is obvious that playing more will generally make you a better player, but the number of games in the FargoRate system is not a good general indicator of playing experience. What makes a player better is dedicated and smart practice outside of league and tournament play. Also, some top players do not have many games in the FargoRate system because they do not compete in many FargoRate-participating tournaments, as is the case with many Asian players who do not travel internationally often. Also, some players with low ratings have many games in the system. And people who have been in the system longer will generally have more games in the system, regardless of what their rating might be.

Physical Advantages of Males Over Females in Pool

Due to muscle physiology differences (stronger and faster muscles), males outperform females in sports involving throwing, jumping, running, lifting, and swimming. The data on this is clear. For example, see the Track and Field World Records in the sidebar. In pool, strength and speed are not critically important factors, but they do make a difference. For the reasons covered in the remainder of the article, males generally have an advantage in pool. These advantages can be especially important at a higher playing level, where very small differences in ability can have a large impact on match outcomes. This is the case with a high level of play in all sports and games, where most top athletes or players have very similar ability levels.

Track and Field
Men and Women World Records

Event	Women	Men
100 Meters	10.49s	9.58s
400 Meters	47.60s	43.03s
800 Meters	1:53.28s	1:42.67
High Jump	2.10m	2.45m
Long Jump	7.52m	8.95m
Shot Put *	22.63m	23.56
Javelin Throw *	72.28m	98.48

*: Note – The implements are lighter for Women, making it easier to throw them.

Below are some of the **physical advantages a male has related to pool**.

A.) Muscle Speed

Males generally have **more strength and faster-twitch muscles** than females, making it easier to execute power shots. The break shot is very important and can often be the deciding factor in a high-level match. Faster break shot speed, with accuracy, generally results in a better ball spread with a greater chance to pocket a ball. Per break statistics from professional tournament matches (see *drdavepoolinfo.com/faq/break/stats/*), pros with good breaks win games more often. The key to success with the break is using as much speed as possible without losing too much accuracy or cue ball (CB) control. Male muscle physiology seems to be better suited to this, as evidenced by top male pro pool players being very effective with break shot accuracy at speeds significantly higher than top pro female pool players. This pattern is also evident in amateur pool leagues.

In the game of 9-ball, where modern tournament break rules (with the 9 racked on the spot, breaking from the box) result in a cut break strategy, power is less of a factor than it once was. The break speeds used in these tournaments is well within the accurate range of top female pros. Maybe we will see females being competitive in World Nine-ball Tour (WNT) events in the future. I certainly hope so.

Other power shots are also important. A good example is a long draw shot, where a low and accurate tip position and significant cue speed are required to position the CB. A typical male with good skills can execute these shots with less effort compared to a typical female. And for any shots requiring fast speed, **if you can generate more cue speed with less effort**, which is the case if you have stronger and faster muscles like males, **you will generally be more accurate and have better control and consistency**. With more effort, there is more of a chance for stroke fundamentals to break down, resulting in the cue not being delivered as straight.

All pool players lose accuracy as they increase effort and cue speed beyond a certain range, especially as they get closer to the maximum speed they can generate. It is simply more difficult to keep everything still and aligned perfectly when you are exerting effort to accelerate the cue. The relationship between shot accuracy and shot speed is also present in other sports like golf, bowling, and tennis. Beyond a certain range of shot speeds, as you use more effort in an attempt to hit or throw a ball with more speed, you will be less accurate with shot direction and distance control. And if you can generate faster speed with less effort, accuracy will be better. Anybody who has played pool or any of these other sports competitively can relate to this effect.

B.) Height

Males are generally taller, which can offer advantages in pool. First, being taller allows you to reach more shots with greater ease and without needing a “mechanical bridge,” use of which often results in less accuracy and control. Second, being taller gives you a better view of the arrangement and angle relationships among the balls on the table. This makes it easier to see the cut angles required to pocket balls. It can also make it easier to see strategic ball-pocketing patterns and be able to visualize CB paths.

However, since many top pro pool players are short, it seems that height is not as big a factor as some people might think. Shane VanBoening (5’11”, middle in left photo), one of the greatest American players of all time, towers over many other pro players and he is not very tall, especially compared to Mike Massey (6’5”, left in left photo) and me (6’3”, right in left photo). Alex Pagulayan (5’3”, right photo next to me) is short and yet he is one of the greatest players of all time.

Being tall is not an indicator of pro pool success

But height also comes with disadvantages in pool. The effort required to bend down and hold a stance on every shot is fatiguing and can lead to many issues. I did a series of videos with a respected sports physical therapist a couple of years ago demonstrating exercises and stretches to help prevent and treat 10 common pool-related physical ailments (see drdavepoolinfo.com/faq/advice/exercise/). I have personally experienced 8 of these 10 ailments over the years, mostly due to my height and the need to bend and twist so much to get low on shots for accuracy.

C.) Hand Size

Males generally have larger hands with longer and stronger fingers, making it easier to elevate the bridge hand with more stability. A taller bridge enables one to more effectively reach over obstacle balls. It also enables a longer stroke length with an elevated bridge, which is particularly helpful with jacked-up shots requiring speed, including jump and power massé shots. But again, as mentioned above, many pro pool players are not very tall and do not have very large hands or long fingers, so any potential disadvantages associated with being short with small hands can apparently be easily overcome.

Other Potential Contributing Factors

There are **many factors that contribute to playing pool at a high level**. Below is a list of some of the things most top players have in common. They:

a. have dedicated much of their life to practicing and playing pool.

b. have tremendous focus and intensity when playing.

c. have developed a wealth of experience and intuition through countless hours of smart practice and successful play.

d. have good eye-hand coordination and can consistently and accurately align and deliver the cue along the desired line with the tip contact point and speed required.

e. have good visual acuity (good eyes or corrected vision) and visual perception, being able to clearly and consistently see the angles of shots and the required lines of aim.

f. have been around, watched, played, and learned from many top players.

g. have very strong desire, dedication, and drive to improve and win.

h. are fearless but are also aware of their limitations.

i. are willing to travel and play often in many tournament or gambling matches against players who will challenge them to their limits and beyond.

j. have the mental and physical stamina necessary to play with excellence over the long hours required in tournament and gambling formats.

Currently, most female players do not have the same opportunities, level of experience, exposure to, or level of ability concerning some of the items above. This is reflected in the FargoRating differences discussed earlier. Beyond the physical advantages described in the previous section, here are some additional possible factors that might contribute to the current performance differences between males and females:

a. **Males generally have more opportunities, experiences, and encouragement during childhood** with activities (sports, physical play, building and taking things apart, video games, etc.) that help improve eye-hand coordination and visualization skills.

b. There have been studies showing that **males generally have better 3D perception and visual-spatial skills** than females. It is not clear if the cause of this is biological or societal; but these skills are critical in pool, especially for aiming, so any differences in this area are important.

c. Males potentially have **evolutionary benefits** from historically being hunters (which required good spatial perception, planning, singular focus, stoicism, fearlessness, etc.) rather than nurturers and gatherers.

d. Many amateur females playing pool are **casual or social players**, and many of them play simply because they have a boyfriend or husband who has a passion for pool. Because these females are likely not as serious about the game, their performance would be expected to be lower.

e. Far **fewer females then males play the sport**. This is due to many potential factors including the lack of female role models and the uninviting atmosphere they often face in the male-dominated and sometimes disrespectful environment of pool halls, where males often:
   - offer unwelcomed and unhelpful “advice” assuming the females do not know how to play,
   - tease them about “breaking like a girl,”
   - “hit” on them, and
   - do other things to make them feel discouraged, uncomfortable, or unwelcome.

Conclusions

Based on the FargoRate data, males currently play pool at a much higher level than females, at all levels of experience. This might in part be due to the number of players involved. Many more males than females play pool, so there will obviously be more who play better. However, males also have slight physical and other possible advantages. Probably the biggest difference, other than societal, is muscle physiology (more fast-twitch muscle fibers in males) allowing for cue speed to be created with less effort, especially with power shots. This potentially allows males to be more accurate and consistent when faster cue speed is used.

Even though women have greater potential to improve by playing against men in more-competitive Open divisions, separate and protected Female divisions are important. A Female division encourages more females to enter and compete, and more females need to compete to attempt to "level the playing field" over time. If females have less likelihood of placing well in an Open division, they will have less incentive to enter. It is not practical to have a separate division for transgender women due to the limited number of participants and the time and costs involved, but both transgender women and biological females are allowed and should be encouraged and supported to compete in Open (Men's) divisions.

Since a transgender woman still has the genes, muscle physiology, societal upbringing, spatial and visual perception skills, and other potential benefits of being born and raised male (especially after undergoing male puberty), she should not be allowed to compete in female-only events based on the data and arguments presented in this article. If the day comes when the level of play of females is on par with males, the question is moot since Female divisions would no longer be required.

I tried to get input from five current top female pros about allowing transgender women to participate in female events. Unfortunately, none of them had an opinion or were willing to share views on a topic so sensitive and controversial in today's world. I don't blame them. If you want to see alternative perspectives on gender in pool, see Mike Page's video and the other content at drdavepoolinfo.com/faq/mental/gender-in-pool/.

Concerning my experience with the trial, see the next page.

PS: I want to thank Mike Page and Bob Jewett who offered excellent suggestions when reviewing an early draft of the article. Thanks guys!

Dr. Dave's Experience with the Case, as an Expert Witness supporting the EBPF

I was hired as an "Expert Witness" based on my background dealing with the physics of pool, my experience teaching and playing the game, and my general knowledge of cue sports. The first task was writing a technical report documenting data and explanations for why males perform at a higher level than females in the game of pool. There were four experts on the case, two on each side dealing with sports biomechanics, physiology, and gender, and two (me included) dealing with the physics of the game. My counterpart was an MIT physicist professor I refer to as "Dr. MIT." Unlike in the US, expert witnesses in the UK are asked to discuss and debate the individual reports and write a joint report summarizing all areas of agreement and disagreement, along with all reasons for differing opinions. I think this is a much better approach than in the US, where all the "dirty laundry" is saved for the trial. Dr. MIT and I had many excellent discussions and debates leading to our joint report. Soon before the case, we both also submitted supplemental reports addressing lingering concerns from the joint report.

The entire process up to this point was relatively relaxed and cordial; although, there was a little stress at times created by "surprise deadlines." But during the trial (April 7-11), the "gloves" came off, and it was time to attack as fiercely as possible. Dr. MIT and I appeared remotely via video conference. Unfortunately for Dr. MIT, I knew a lot more about pool than he did, and this became a problem for him and his side during testimony. The lead attorney on my side did a good job "exposing his vulnerabilities," making some of his work look like "rubbish."

The time zone difference between Colorado and London is significant, so I had to be online at 3am two days in a row, ready to carefully listen and take notes during testimony by the other experts. And on the second day, I had to endure 3 straight hours of questioning, mostly cross-examination by the claimant attorney with continual attempts to discredit or attack things I had written before the trial or said earlier in the testimony. It felt like enemy interrogation, and I had to be extremely careful with everything I said, attempting to anticipate what attack might be disguised in a sometimes-innocent-sounding question. Regardless, I very much enjoyed the entire expert witness process from start to finish, and I hope I get a chance to do it again in the future.

The only thing I didn't like about the experience is we will not know the verdict for up to 2 months. The judge needs to review mountains of reports and evidence, review all pertinent law associated with the discrimination claims, and take everything (including all the evidence and expert testimony presented in court) into consideration before coming to his final verdict. I expect my side (the EBPF) will come out on top and not be found guilty of discrimination by excluding Harriet Haynes from competing against biological females in the Ladies division, but you never know which way a judge might lean in such controversial cases. I think I did my best to explain and present data on why males (and transgender females) have a significant advantage, mostly with the break shot.